

Derivation of the Birthday Paradox Formula

Probability of Collisions in Randomly Chosen Outcomes

To derive the Birthday Paradox formula, we calculate the probability that in a set of k randomly chosen items (e.g., people) with N possible equally likely outcomes (e.g., $N = 365$ days in a year), at least two items share the same outcome.

Step 1: Exact Probability Formulation

Calculating the probability of "at least one match" directly is complex because it includes 2-way, 3-way, or multiple pair overlaps. It is much easier to use the complement rule:

$$P(\text{Match}) = 1 - P(\text{No Match})$$

For $P(\text{No Match})$, we select items one by one such that each new item has a unique value:

1. 1st item: Can have any value $\Rightarrow N/N$
2. 2nd item: Must differ from the 1st item $\Rightarrow (N - 1)/N$
3. 3rd item: Must differ from the 1st and 2nd items $\Rightarrow (N - 2)/N$
4. i -th item: Must differ from the previous $i - 1$ items $\Rightarrow [N - (i - 1)]/N$

Multiplying these independent choices for all k items gives:

$$\begin{aligned} P(\text{No Match}) &= \prod_{i=0}^{k-1} [(N - i)/N] \\ &= (N/N) \times ((N-1)/N) \times ((N-2)/N) \times \dots \times ((N-k+1)/N) \end{aligned}$$

We can rewrite each term in the product as $(1 - i/N)$:

$$P(\text{No Match}) = \prod_{i=0}^{k-1} (1 - i/N)$$

Thus, the exact match probability is:

$$P(\text{Match}) = 1 - \prod_{i=0}^{k-1} (1 - i/N)$$

Step 2: Applying Taylor Series Approximation

For small values of x (specifically $x \ll 1$), the Taylor series expansion of $e^{(-x)}$ is:

$$e^{(-x)} = 1 - x + x^2/2! - \dots \approx 1 - x$$

Setting $x = i/N$, since $k \ll N$, i/N is very small, allowing us to approximate:

$$1 - i/N \approx e^{(-i/N)}$$

Substitute this approximation back into the product for $P(\text{No Match})$:

$$P(\text{No Match}) \approx \prod_{i=0}^{k-1} e^{(-i/N)}$$

Using exponential properties ($\prod e^{(a_i)} = e^{(\sum a_i)}$):

$$\begin{aligned} P(\text{No Match}) &\approx e^{[-\sum_{i=0}^{k-1} i/N]} \\ &= e^{[-(1/N) \sum_{i=0}^{k-1} i]} \end{aligned}$$

Step 3: Simplifying the Summation

The summation $\sum_{i=0}^{k-1} i$ is an arithmetic series $(0 + 1 + 2 + \dots + (k - 1))$:

$$\sum_{i=0}^{k-1} i = (k - 1)k/2 = (k^2 - k)/2$$

For large k , $k^2 \gg k$, so we can approximate $k^2 - k \approx k^2$:

$$\sum_{i=0}^{k-1} i \approx k^2/2$$

Substituting this back into our exponential expression:

$$P(\text{No Match}) \approx e^{(-k^2/(2N))}$$

Therefore, the probability of at least one collision/match is:

$$P(\text{Match}) \approx 1 - e^{(-k^2/(2N))}$$

Step 4: Deriving the 50% Probability Threshold ($k \approx 1.177\sqrt{N}$)

To find the number of items k required to achieve a 50% probability of a match ($P(\text{Match}) = 0.5$):

$$0.5 = 1 - e^{(-k^2/(2N))}$$

Rearranging:

$$e^{(-k^2/(2N))} = 0.5$$

Take the natural logarithm ($\ln$) of both sides:

$$-k^2/(2N) = \ln(0.5) = -\ln(2)$$

Multiply both sides by -1 :

$$k^2/(2N) = \ln(2)$$

Solve for k^2 :

$$k^2 = 2N \ln(2)$$

Take the square root of both sides:

$$k = \sqrt{2N \ln(2)} = \sqrt{2 \ln(2)} \times \sqrt{N}$$

Now compute the numerical constant $\sqrt{2 \ln(2)}$:

- $\ln(2) \approx 0.693147$
- $2 \ln(2) \approx 1.386294$
- $\sqrt{1.386294} \approx 1.17741$

Substituting this value back yields the final bound:

$$k \approx 1.177\sqrt{N}$$

Summary Example ($N = 365$)

For standard 365 days in a year:

$$k \approx 1.177\sqrt{365} \approx 1.177 \times 19.105 \approx 22.49 \approx 23 \text{ people}$$

This proves that with just 23 people, there is a $\geq 50\%$ chance that two people share a birthday.

In cryptography (where $N = 2^n$ for an n-bit hash output), this derivation formally explains why finding collisions only requires:

$$k \approx 1.177\sqrt{(2^n)} = O(2^{(n/2)}) \text{ operations}$$